

AGML Lab: Week 2

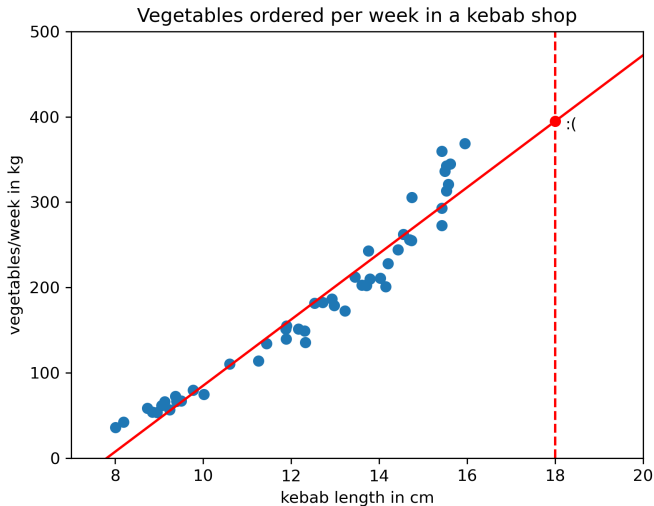
Polynomial Regression



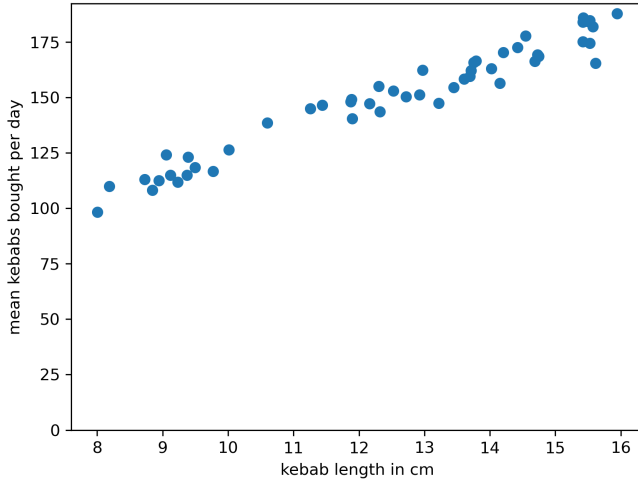
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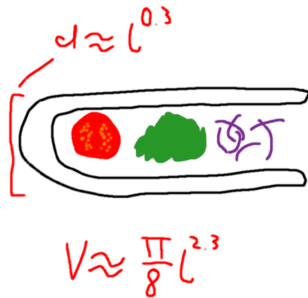
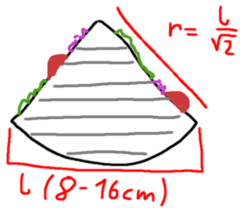
29.04.2024

(Jupyter Notebook)



Longer kebabs are bought more





$$f(x) = a_dx^d + \dots + a_2x^2 + a_1x + a_0$$

$$X = \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ x_1 & x_2 & x_3 & \dots & x_m \\ x_1^2 & x_2^2 & x_3^2 & \dots & x_m^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_1^d & x_2^d & x_3^d & \dots & x_m^d \end{pmatrix} \quad \theta = \begin{pmatrix} a_0 \\ \vdots \\ a_d \end{pmatrix}$$

$$y = X^\top \theta + \epsilon$$

I feel like i have seen this equation before ...

We just have to solve the normal equations again, and maybe sprinkle in some regularization this time:

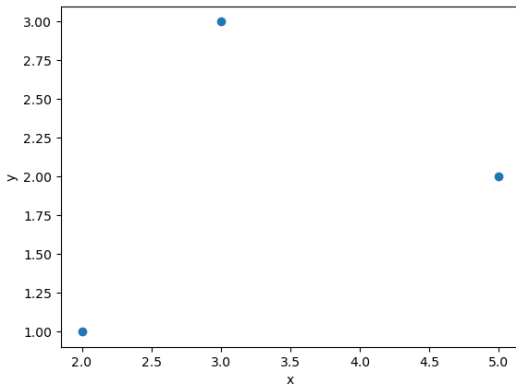
$$\left(XX^{\top} + \lambda\mathbb{1}\right)\hat{\theta} = Xy$$

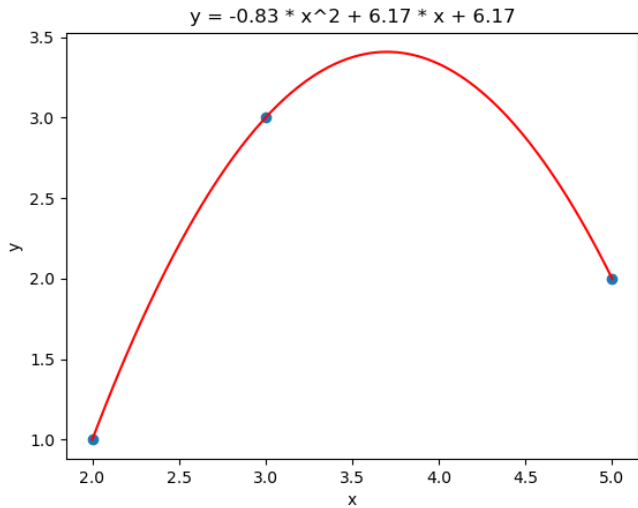


(Jupyter Notebook)

Toy Dataset:

$$D = \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 3 \end{pmatrix}, \begin{pmatrix} 5 \\ 2 \end{pmatrix} \right\}$$





1. Why do we regularize?
 2. How big should λ be?
 3. What do you think happens to the predictions if we increase λ too much?
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